Jaume de Dios Pont

Curriculum Vitae

60 5th Avenue
New York, NY 10011
✉ *jdedios@nyu.edu*

Employment

2026 – **NYU Center for Data Science**, *CDS Faculty Fellow*.

2023 – 2025 **ETH Zurich**, *Postdoctoral Researcher*, Mentor: Svitlana Mayboroda. Simons Collaboration on Localization of Waves.

May '23 – Sep '23 **Microsoft Research**, *Research Intern*, Physics of AGI group. Mentors: Jerry Li; Adil Salim.

2020 – 2022 **UCLA**, *Teaching Assistant*.

Education

2023 **PhD Mathematics**, *UCLA*.
Part I: Uniform Estimates for Operators Involving Polynomial Curves. Part II: Decoupling Estimates for Fractal and Product Sets. Supervised by T. Tao

2018 **MSc Mathematics**, *ETH Zurich*, (Grade: 5.76/6).
Thesis: Quantum Loewner Evolution (E. Powell and W. Werner)

2017 **BSc Mathematics**, *Universitat Autònoma de Barcelona*, (#1 Rank, Grade: 9.71/10).
Thesis: Oscillatory integrals and the Kakeya Conjecture (J. Garnett; J. Verdera)

2017 **BSc Physics**, *Universitat Autònoma de Barcelona*, (#1 Rank, Grade: 9.62/10).
Thesis: KCM-related experiments (F.X. Alvarez; A. Lopeandia)

Postgraduate Awards and Scholarships

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Papers and Preprints

[1] **Cantor measures with odd base do not admit Fourier frames**, ***de Dios Pont, J.***, *Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2607.08656.

[2] **Stable Phase Retrieval for Spans of Independent Random Variables**, *Pedro Abdalla,* ***de Dios Pont, J.***, *João P. G. Ramos & Mitchell A. Taylor*, Preprint (2026), arXiv:2607.06693.

[3] **Geometric Dyson Brownian Motions and the Free Log-Normal Limit for a Non-Square Gaussian Matrix Product**, *Mufan Li,* ***de Dios Pont, J.***, *Mihai Nica & Daniel M. Roy*, Preprint (2026), arXiv:2606.30831.

[4] **On the existence problem of regular Gabor frames**, ***de Dios Pont, J.***, *Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2606.26052.

[5] **L^2-Stability for STFT phase retrieval**, *Susanna Bertolini,* ***de Dios Pont, J.***, *Ben Pineau, Mitchell A. Taylor & João P. G. Ramos*, Preprint (2026), arXiv:2605.20527.

[6] **Sharp bounds on the failure of the hot spots conjecture**, ***de Dios Pont, J.***, *Alexander W. Hsu & Mitchell A. Taylor*, Preprint (2025), arXiv:2508.16321.

[7] **Convex sets can have interior hot spots**, ***de Dios Pont, J.***, Preprint (2024), arXiv:2412.06344.

[8] **Predicting quantum channels over general product distributions**, *Sitan Chen*, ***de Dios Pont, J.***, *Jun-Ting Hsieh, Hsin-Yuan Huang, Jane Lange & Jerry Li*, Proceedings of the 38th Conference on Learning Theory (COLT), PMLR 291 (2025), 986–1007, arXiv:2409.03684.

[9] **Periodicity and decidability of translational tilings by rational polygonal sets**, ***de Dios Pont, J.***, *Jan Grebík, Rachel Greenfeld & José Madrid*, Expositiones Mathematicae 44(1) (2026), article 125620, arXiv:2408.02151.

[10] **A new proof of the description of the convex hull of space curves with totally positive torsion**, ***de Dios Pont, J.***, *Paata Ivanisvili & José Madrid*, Michigan Mathematical Journal, advance publication (2025), 1–47, arXiv:2201.12932.

[11] **Query lower bounds for log-concave sampling**, *Sinho Chewi*, ***de Dios Pont, J.***, *Jerry Li, Chen Lu & Shyam Narayanan*, Journal of the ACM 71(4) (2024), article 29, 1–42; conference version: FOCS 2023, 2139–2148, arXiv:2304.02599.

[12] **Uniform Fourier Restriction Estimate for Simple Curves of Bounded Frequency**, ***de Dios Pont, J.***& *Helge Jørgen Samuelsen*, The Journal of Geometric Analysis 36(6) (2026), article 209, arXiv:2303.11693.

[13] **Additive energies on discrete cubes**, ***de Dios Pont, J.***, *Rachel Greenfeld, Paata Ivanisvili & José Madrid*, Discrete Analysis 2023:13, 16 pp., arXiv:2112.09352.

[14] **Decoupling for fractal subsets of the parabola**, *Alan Chang*, ***de Dios Pont, J.***, *Rachel Greenfeld, Asgar Jamneshan, Zane Kun Li & José Madrid*, Mathematische Zeitschrift 301(2) (2022), 1851–1879, arXiv:2012.11458.

[15] **Role Detection in Bicycle-Sharing Networks Using Multilayer Stochastic Block Models**, *Jane Carlen*, ***de Dios Pont, J.***, *Cassidy Mentus, Shyr-Shea Chang, Stephanie Wang & Mason A. Porter*, Network Science 10(1) (2022), 46–81, arXiv:1908.09440.

[16] **On classical inequalities for autocorrelations and autoconvolutions**, ***de Dios Pont, J.***& *José Madrid*, Preprint (2021), arXiv:2106.13873.

[17] **On Sparsity in Overparametrised Shallow ReLU Networks**, ***de Dios Pont, J.***& *Joan Bruna*, Preprint (2020), arXiv:2006.10225.

[18] **A geometric lemma for complex polynomial curves with applications in Fourier restriction theory**, ***de Dios Pont, J.***, Preprint (2020), arXiv:2003.14140.

[19] **Translational tilings by axis-parallel polygonal sets**, ***de Dios Pont, J.***, *Jan Grebík, Rachel Greenfeld & José Madrid*, Preprint (2026), arXiv:2610.15785.

[20] **Phase retrieval from a uniformly discrete point set**, ***de Dios Pont, J.***, *Josef Greilhuber, Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2609.28341.

[21] **Banach lattices and phase retrieval: A case study for the use of AI in mathematics**, ***de Dios Pont, J.***, *Lukas Liehr, David Muñoz-Lahoz, Mitchell A. Taylor & Pedro Tradacete*, Preprint (2026), arXiv:2608.07396.

[22] **Counterexamples to Schiffer's Conjecture**, *Gonzalo Cao-Labora &* ***de Dios Pont, J.***, Preprint (2026), arXiv:2608.05114.

[23] **Gabor Frames of Totally Positive Functions: A Complete Characterization**, ***de Dios Pont, J.***, *Karlheinz Gröchenig, Lukas Liehr, Irina Shafkulovska & Mitchell A. Taylor*, Preprint (2026), arXiv:2608.04992.

[24] **Almost-Orthogonality in Lp Spaces: A Case Study with Grok**, *Ziang Chen*, ***de Dios Pont, J.***, *Paata Ivanisvili, José Madrid & Haozhu Wang*, Preprint (2026), arXiv:2605.05192.

[25] **Lea: An Agent Backbone for Mathematician-Controlled Formalization in Lean**, *S. Patel, D. Arturi, K. Gomez,* ***de Dios Pont, J.***, *J. Poco, E. Teixeira, S. Westrick, C. Silva & C. Hegde*, NeurIPS 2026 Workshop on Mathematical Reasoning and AI (MATH-AI), forthcoming.

Talks

Research talks

Spectral theory and the hot spots conjecture

- 2026 Simons Collaboration on Localization of Waves Annual Meeting (Feb '26)
- NYU CDS MaD Seminar (Feb '26)
- Joint Mathematics Meetings 2026 (Washington, D.C.) (Jan '26)
- Instituto de Ciencias Matemáticas (ICMAT) Seminar (Sep '25)
- ISM Discovery School — Interactions between Convex Geometry and Spectral Analysis (Montreal) (Jul '25)
- UK Spectral Theory Network Workshop (University of Reading) (Aug '25)
- Workshop on Spectral Geometry, PDEs and Mathematical Physics (FernUni Hagen) (Jul '25)
- Fourier Analysis and Beyond I (IMPA, Rio de Janeiro) (Jul '25)
- ETH Zürich Random Walks Group meeting (Yuansi Chen) (May '25)
- LSEC Seminar (Apr '25)
- University of Edinburgh Analysis Seminar (Mar '25)
- Lehigh University Geometry Seminar (Mar '25)
- Institut Camille Jordan Analysis Seminar (Lyon) (Mar '25)
- Spectral Geometry in the Clouds (Mar '25)
- Simons Collaboration on Localization of Waves Annual Meeting — Poster Session (Flatiron Institute) (Feb '25)
- Virginia Tech Analysis Seminar (Feb '25)
- Seminari d'Anàlisi UB-UAB (Jan '25)
- ETHZ Analysis Seminar (Sep '24)
- Hausdorff Center for Mathematics Colloquium (Bonn) (Oct '24)
- 2024 Simons Collaboration on Localization of Waves Meeting (Oct '24)
- NYU Convexity Seminar (May '26)
- Brown University PDE Seminar (May '26)
- South China University of Technology (Jun '26)
- University of Macau (Jun '26)
- University of Macau (Jun '26)

Lower bounds for sampling

- CRM — Mathematical Foundations of Machine Learning (Barcelona) (Jan '26)
- Hausdorff Research Institute for Mathematics — Boolean Analysis in Computer Science (HIM, Bonn) (Oct '24)
- BIRS-IMAG Workshop (Granada) (Jun '24)
- UCLA Analysis Seminar (May '24)
- HIM Winter School on Geometry and Probability (Jan '24)
- University of Rochester Computer Science Seminar (May '23)
- NYU Courant Analysis Seminar (Mar '23)
- Microsoft Research Theory Seminar (Dec '22)
- NYU MaD Group Meeting (Dec '22)

Exploratory mathematics with LLM assistants

- MPS Workshop on Computation in Mathematics (Flatiron Institute) (Feb '25)
- La Salle–Universitat Ramon Llull, Applied Mathematics research group (Jun '26)
- Universitat Autonoma de Barcelona (Jun '26)
- Mini Symposium on AI for Mathematics, University of Macau (Jun '26)
- Mini Symposium on AI for Mathematics, University of Macau ()

Uniformity for polynomial curves

- Rutgers University Analysis Seminar (Oct '23)
- University of Rochester Combinatorics Seminar (May '23)
- University of Minnesota PDE Seminar (Sep '22)
- Bilbao Analysis and PDE Seminar (BCAM) (Mar '22)
- UAB Analysis Seminar (Universitat Autònoma de Barcelona) (Mar '22)
- UK Virtual Harmonic Analysis Seminar (Fourier 2.0) (Oct '21)
- Bristol Analysis and Geometry Seminar (Feb '22)
- Stanford University Analysis and PDE Seminar (Oct '22)

Decoupling for Cantor sets

- Harmonic Analysis and Fractal Sets Conference (HAFS, Columbus OH) (Mar '23)
- Fourier Restriction Online 2021 (Mar '21)

Uniform boundedness for certain operators parametrized by polynomial curves

- UW Madison Analysis Seminar (Nov '22)
- Harmonic Analysis on Manifolds Summer School (UW Madison) (Aug '22)
- ETHZ Analysis Seminar (Mar '22)
- Probability and Analysis Webinar (PAW) (Aug '21)
- UC Davis Student-Run Analysis and PDE Seminar (Feb '21)
- Seminari d'Anàlisi UB-UAB (Nov '20)
- Online Analysis Research Seminar (OARS) (Dec '20)
- NTNU Analysis and PDE Seminar (Oct '23)

A Function Space Perspective for Regularised and Overparametrised Shallow ReLU Networks

- NYU, MaD Group Meeting (Oct '20)

Recent progress on the hot spots conjecture

- COST mSPACE Kick-off Meeting (Milan) (Mar '26)
- Isaac Newton Institute — Geometric Spectral Theory and Applications (Feb '26)
- Institute for Advanced Study, Analysis and Mathematical Physics (Nov '26)

Finding counterexamples to Pompeiu's and Schiffer's conjectures

- NYU Courant Analysis Seminar (Oct '26)
- Drexel University (Oct '26)

The Future of AI and Math

- Math & AI Summer in Seattle, Amazon Seattle Campus (Aug '26)
- Organizing Mathematical Knowledge in the Age of AI and Formalization, National Academy of Sciences (Jul '26)

Mathematics in the age of AI

- CRM / BGSMath (Dec '26)
- Joint Brown–Yale seminar, Yale University (Apr '26)
- Yale Institute for Foundations of Data Science (Apr '26)

Expository talks

- **Generació de variables aleatòries**, Valentia Matemàtica Summer School (Jun '25)
- **Power-type cancellation for the simplex Hilbert transform**, Kopp Summer School Reading Group (Bonn) (Sep '23)
- **Decoupling and applications: from PDEs to Number Theory**, Microsoft Research Theory Seminar (Jun '23)
- **Localization of eigenfunctions via an effective potential**, Kopp Summer School Reading Group (Bonn) (Oct '22)
- **A proof of the Sensitivity Conjecture (after H. Huang)**, AIM Workshop: Analysis on the Hypercube with Applications to Quantum Computing (Jun '22)
- **Euclidean Forward-Reverse Brascamp-Lieb Inequalities**, Brascamp-Lieb Summer School Reading Group (Kopp, Germany) (Sep '21)
- **A proof of the sensitivity conjecture**, UCLA Participating Analysis Seminar (Reading Group) (Nov '21)
- **Decoupling and applications: from PDEs to Number Theory**, SIMBa Seminar (UB / BGSMATH) (Oct '20)
- **Mathematics for phase retrieval**, Paul Scherrer Institut (Mar '26)
- **The sensitivity theorem**, Stanford Kiddie Colloquium (Oct '22)
- **Duminil-Copin's work and the Ising model**, UCLA GSO (May '23)
- **You already know how to read Lean (a very elementary primer to Lean)**, Aspen Meeting on Foundation Models ()

Talks are grouped by topic, even when the covered material changed between instances.

Research Visits (> 1 week)

Jul '25 **New York**, *Flatiron Institute*.

Oct '24 **Hausdorff Institute of Mathematics, Bonn**, *Boolean Analysis in Computer Science*.

Jan '24 **Hausdorff Institute of Mathematics, Bonn**, *Dual Trimester Program: Synergies between modern probability, geometric analysis and stochastic geometry*.

Oct '23 – Nov '23 **NTNU, Trondheim**, *NTNU visit*.

May '23 – Sep '23 **Seattle, WA**, *Microsoft Research Internship*.

Oct '22 **Palo Alto, California**, *Stanford University*.

Sep '22 **UMN, Minneapolis**, *University of Minneapolis*.

Jun '22 – Jul '22 **ICMS, Edinburgh**, *Fourier Analysis @200*.

Jan '22 – Mar '22 **Hausdorff Institute of Mathematics, Bonn**, *Interactions between Geometric measure theory, Singular integrals, and PDE*.

Aug '21 **Hausdorff Mathematical Institute, Bonn**, *Harmonic Analysis and Analytic Number Theory, Dual trimester program*.

May '21 – Jun '21 **Hausdorff Mathematical Institute, Bonn**, *Harmonic Analysis and Analytic Number Theory, Dual trimester program*.

Feb '26 – Mar '26 **Cambridge, UK**, *Isaac Newton Institute - Geometric Spectral theory and Applications*.

Teaching Experience

ETH Zurich (Main Instructor)

- **Formalizing Mathematics in Lean** Spring 2025; Autumn 2025

ETH Zurich (Teaching Assistant)

- **Differential Geometry** Spring 2024
- **Differential Geometry I** Autumn 2023

UCLA (Teaching Assistant)

- **Math 131A (Real Analysis)** Fall 2020; Spring 2022
- **Math 134 (Linear and Nonlinear Systems of Differential Equations)** Fall 2021
- **Math 135 (Ordinary Differential Equations)** Fall 2021
- **Math 33B (Differential Equations)** Spring 2021
- **Math 32A (Calculus of Several Variables)** Fall 2020; Winter 2021
- **Math 32B (Calculus of Several Variables)** Winter 2021

NYU Courant (Instructor)

- **Advanced Topics in Analysis (MATH-GA.2650-001)** Fall 2026

NYU Center for Data Science (Course Advisor)

- **Capstone Continuation** Spring 2026

Reviewing

Reviewer for: PA^{2}IN — PDE, Analysis, Artificial Intelligence, and Numerics, IMPA, Rio de Janeiro, 25–29 January 2027 (upcoming), Phase Retrieval: Connecting Theory and Applications, RWTH Aachen University, 21–25 September, Palomar, Mathematics in the Age of AI Seminar, NYU Courant, OpenEvolve hacking/learning session, Institute for Theoretical Studies, ETH Zürich, Chalk & Cheese, Institute for Theoretical Studies, ETH Zürich.

Misc. Coding Skills

- Proficient in Python (incl. Jax, NumPy/SciPy/Matplotlib)
- Git/Version Control
- LaTeX

Undergraduate Research Experience

2017 **GNAM**, *Grup de Nanomaterials, UAB*.
Research on heat conduction beyond the Fourier equations at the nanoscale, with a focus on the KCM diffusion model.

2016 **ICFO**, *Summer Fellowship of the Institute of Photonic Sciences*.
Research fellow in the group of Antonio Acín (Quantum Information). Proved impossibility results on the creation of superpositions of unknown quantum states. Supervisor: Dr. Michał Oszmaniec.

2015 **The Dark Energy Survey Project**, *IFAE – Institute for High Energy Physics*.
Study of theoretical models regarding the harmonic spectrum of galaxy density distributions. Supervisor: Dr. Ramon Miquel.